

COVIZ: Visualization of Effects of COVID-19 on New York City Through Socially Impactful Virtual Reality

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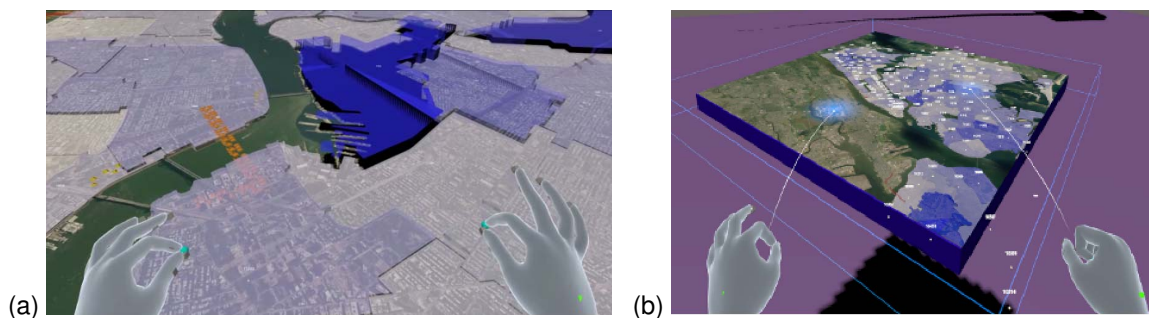


Figure 1: (a) View of 3D map showing height fields representing infection rates along with color-coded vehicular traffic rates. (b) User scales 3D map using point-and-pinch gestures.

ABSTRACT

This work is the product of a collaboration between students studying computer science and social work to visualize the impacts and effects of COVID-19 in New York City in a virtual environment (VE). As a proof of concept, the team chose two datasets from NYC Open Data; COVID-19 infection cases and rates per zip code and vehicular traffic rates within the five boroughs of New York City. To foster unexplored insights into the relationship between these data, we developed a virtual reality application that provides a stronger sense of embodiment and ownership of urban visualization analysis when manipulating 3D virtual maps for comparison in a VE.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Visualization—Visualization application domains—Information visualization

1 INTRODUCTION

Searching for meaningful relationships between data through visualization can be tedious, even ambiguous, on a monitor or computer screen. The rising accessibility of virtual reality may make possible less obvious insights into such relationships, paving the way for a new era of scientific tools to benefit future research in an array of domains. We propose that exploring such data in a virtual environment (VE), where users can directly manipulate visualized data in multiple dimensions through manual interaction, may not only be a more pleasurable experience, but may further understanding of the data.

The data aggregated from the COVID-19 pandemic in New York City provides a wealth of opportunities for obtaining new insights,

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correlations, and discrepancies amongst different demographics and neighborhoods, potentially allowing for us to learn how to better mitigate the spread of future pandemics.

We are developing a system in which multiple datasets from different sources can be combined onto a single 3D map so that a user may explore and draw conclusions with an easy and intuitive user interface. Users can then compare different time slices (we have chosen a per-week basis for both COVID-19 infection rates and vehicular traffic rates) on separate map instances that they place side by side for comparative analysis.

Our work also emphasizes usability through bare-hand control of objects in the VE. Eliminating controllers fosters a higher degree of portability and a simpler user interface that eliminates the need to memorize button and joystick mappings. Students who have utilized the current early version of our system find exploring the data in such a VE provides a more compelling experience than the desktop systems they have used, while increasing overall engagement of data analysis. One student user remarked, “it’s a fantastic immersive tool to help educators, students and learners come to new conclusions about the data they’re exploring. I had no idea you could do this with data and it makes me want to use the application more.”

2 RELATED WORK

Visualizing data in a VE poses the risk of making it difficult to understand or even being occluded by other data. Approaches to negotiate such hurdles have already been investigated. For example, Selective Dynamic Manipulation (SDM) [1] developed a framework to present and analyze large and diverse datasets. Displaying data in a way that doesn’t confuse or frustrate the user is an art in and of itself. Interacting with data makes the task more difficult. Systems such as ImAxes [2] demonstrate innovative and effective techniques that take advantage of immersive 3D display and manipulation. We are inspired by such research to try to develop visualizations that will not overwhelm or frustrate users.

3 IMPLEMENTATION

3.1 Design

Our system is designed for the Oculus Quest using Unity (<https://unity.com>) 3D and the Mapbox SDK (<https://www.mapbox.com>). To render accurate geographic data, the system interactively queries Mapbox for the proper coordinates. We couple this data with the longitude and latitude attributes of two datasets from NYC Open Data: COVID-19 infection cases and rates per zip code¹ and vehicular traffic rates within the five boroughs of New York City².

3.2 Visualization

Since the COVID-19 dataset is partitioned by zip codes, we felt it appropriate to visualize it by overlaying data above the map plane by their geographic coordinates in a 3D height-field representation with regions of varying shades of blue and height (darker and taller for higher infection rates). As shown in Fig. 1(a), zip code numbers are centered above each partition for convenient reference. This makes vehicular traffic data easier to identify, represented above the COVID-19 infection rate data for all major expressways in the form of moving colored blocks. The color and speed of the moving blocks indicate the level of traffic (red and slow: maximum congestion, orange and medium speed: slight congestion, yellow and medium speed: average congestion, green and fast: low congestion).

3.3 User Interaction

We felt it was important to design the system such that the user only needs to use their hands to interact: less hardware and less stress (for both users and developers). This strategy also makes it convenient for the user to rest their hands (e.g., on their hips or with their arms crossed) so that they can focus on analyzing the data and potentially work for longer periods of time within the VE.

To provide a more intuitive experience, we integrated Microsoft Mixed Reality Toolkit (<https://docs.microsoft.com/en-us/windows/mixed-reality/develop/unity/mrtk-getting-started>) for hands-only interactions: users can translate and rotate each map plane with a one-handed point-and-pinch technique (providing cubic bounding boxes with corner indicators for manipulations) or scale it using both hands to point-and-pinch, as shown in Fig. 1(b).

When the user places either hand in an open palm-up pose in their field of view, a hand-interaction menu (Fig. 2) provides additional options through three buttons that enable adding a new map to the environment, or moving forward or backward on the selected map by one week. We found this implementation useful and intuitive, allowing the user to focus on the data they are exploring.

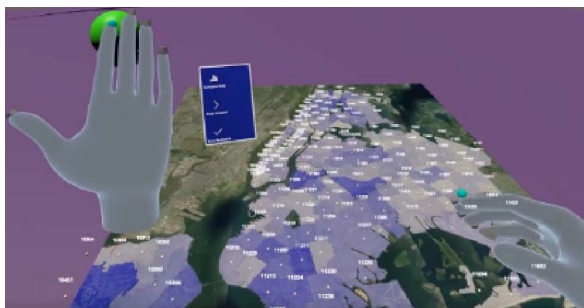


Figure 2: Hand interaction menu.

¹<https://data.cityofnewyork.us/Health/COVID-19-Daily-Counts-of-Cases-Hospitalizations-an/rc75-m7u3>

²<https://data.cityofnewyork.us/Transportation/DOT-Traffic-Speeds-NBE/i4gi-tjb9>



Figure 3: Comparing COVID-19 infection rates between two zip codes.

When a user wants to investigate a particular zip code, they may select or deselect it by pushing on a small sphere that appears below the zip code. This causes a floating billboard to be displayed showing detailed COVID-19 infection rate data for that zip code (Fig. 3). This is particularly useful when comparing data between zip codes. One user of the prototype application noted that, "the immersive environment provides a new sense of agency over the data. It's empowering to take control of the data in a way that I didn't have with a computer screen."

4 FUTURE WORK

We believe this system has the potential to aid users in understanding the differential effects of COVID-19 on local communities and we are working on extending our early prototype. Future iterations will include:

- Adding additional datasets to better understand people's behavior during the pandemic in New York City.
- Importing datasets directly into Unity to decrease initial render time.
- Supporting collaboration between multiple simultaneous users.

5 CONCLUSIONS AND DISCUSSION

We are creating a 3D visualization tool in a VE to explore datasets related to the impact and effects of the COVID-19 pandemic on New York City. Such a tool can aid in research on the socioeconomic disparities between neighborhoods. Our system is designed so that additional datasets can be integrated, providing a more robust interpretation of correlations involving COVID-19 spread and infection rates.

User interactions were designed carefully, focusing primarily on ease of use and accessibility. Our team sees the potential for this system not only to be used as a scientific tool for comparative analyses, but also to be introduced into classrooms so that students may better comprehend the magnitude of the pandemic.

REFERENCES

- [1] M. C. Chuah, S. F. Roth, J. Mattis, and J. Kolojejchick. SDM: Selective dynamic manipulation of visualizations. In *Proceedings of the 8th Annual ACM Symposium on User Interface and Software Technology*, UIST '95, p. 61–70. Association for Computing Machinery, New York, NY, USA, 1995. doi: 10.1145/215585.215654
- [2] M. Cordeil, A. Cunningham, T. Dwyer, B. H. Thomas, and K. Marriott. ImAxes: Immersive axes as embodied affordances for interactive multivariate data visualisation. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology*, UIST '17, p. 71–83. Association for Computing Machinery, New York, NY, USA, 2017. doi: 10.1145/3126594.3126613